

A three-coordinate Ag(I) complex based on the V-shaped ligand 1,3-bis(1-ethylbenzimidazol-2-yl)-2-thiapropane: synthesis, crystal structure and DNA-binding properties

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The complex [Ag(bebt)(cinnamate)] [bebt = 1,3-bis(1-ethylbenzimidazol-2-yl)-2-thiapropane], has been synthesised and characterised by physico-chemical and spectroscopic methods. Single-crystal X-ray diffraction revealed that the coordination environment of the Ag(I) complex can be described as trigonal planar, and the Ag(I) atom is coordinated by one oxygen atom from a cinnamate anion and two nitrogen atoms from ligand bebt. Experimental studies of the DNA-binding properties indicated that the free ligand and the complex bind to DNA via the intercalation mode, and the binding affinity of the complex was found to be greater than that of just bebt.

Keywords: benzimidazole, cinnamate, Ag(I) complex, crystal structure, DNA-binding

The properties of benzimidazole and its derivatives have been studied under a broad spectrum of biological activities for many years. The benzimidazole moiety is found in various biologically active compounds having antiviral, antiulcer, antihypertensive, anticancer, and antihistaminic properties.^{1–5} Bis-benzimidazole ligands, which represent a class of aromatic N–donor organic linkers, are less developed.^{6–8} DNA-interactive bis-benzimidazole analogues such as Hoechst 33258 have also been prepared to explore the potential for anticancer activity. Studies on the interaction of transition metal complexes with DNA continue to attract the attention of researchers due to their importance in design and development of synthetic restriction enzymes, chemotherapeutic drugs and DNA foot-printing agents.^{9–16} In this respect, silver(I) complexes have attracted a great deal of attention due to their biological and pharmacological properties which provide them with antibacterial and antifungal activities.^{17–22} The study of Ag(I) complexes containing benzimidazole-based ligands has received a great deal of attention during the past decades not only owing to their rich coordination chemistry but also due to probable applications in cancer therapy, molecular biology and photodynamic therapy (PDT).^{23–27}

We have investigated a series of V-shaped bis-benzimidazole ligands and their complexes.^{28,29} In the study reported here, a new ligand 1,3-bis(1-ethylbenzimidazol-2-yl)-2-thiapropane (bebt) and its Ag(I) complex have been synthesised and the DNA-binding properties of the complex investigated.

Experimental

C, H and N elemental analysis was determined using a Carlo Erba 1106 elemental analyser. IR spectra were recorded on a Nicolet FT-VERTEX 70 spectrometer in the range 4000–400 cm^{–1} using KBr pellets. Fluorescence spectra were recorded on a LS-45 spectrofluorophotometer. Electronic spectra were taken on a Lab-Tech UV Bluestar spectrophotometer. Electrolytic conductance measurements were made with a DDS-11A type conductivity

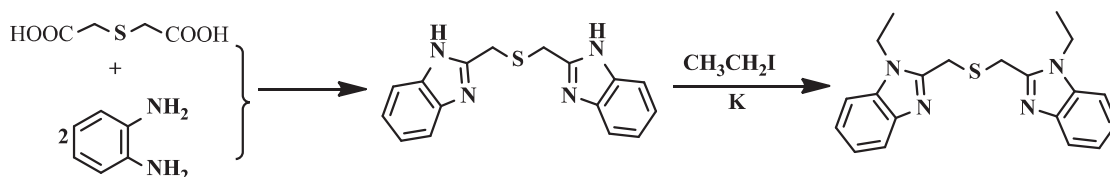
bridge using 10^{–3} mol dm^{–3} solutions in DMF at room temperature. ¹H NMR spectra were obtained with a Mercury plus 400 MHz NMR spectrometer with TMS as internal standard and CDCl₃ as solvent.

Calf thymus DNA (CT-DNA) and ethidium bromide (EB) were obtained from Sigma-Aldrich. Tris-HCl buffer was prepared using doubly-distilled water. Silver cinnamate was prepared by reaction of silver nitrate and sodium cinnamate. Other reagents and solvents were reagent grade obtained from commercial sources and used without further purification. The solution of CT-DNA gave a ratio of UV-Vis absorbance at 260 and 280 nm, A_{260}/A_{280} , of 1.8 – 1.9, indicating that the DNA was sufficiently free of protein.³⁰ The stock solution of the complex was dissolved in DMF at the concentration 3 × 10^{–3} M. The stock solution of DNA (2.5 × 10^{–3} M) was prepared in 5 mM Tris-HCl/50 mM NaCl buffer (pH = 7.2), stored at 4 °C and used within 4 d. The DNA concentration was determined by measuring the UV-Vis absorption at 260 nm, taking the molar absorption coefficient (ϵ_{260}) of CT-DNA as 6600 M^{–1} cm^{–1}.³¹

The synthetic route to ligand bebt is shown in Scheme 1.

1,3-Bis(1-ethylbenzimidazol-2-yl)-2-thiapropane (bebt) (1): 1,3-Bis(benzimidazol-2-yl)-2-thiapropane (5.88 g, 0.02 mol), synthesised by a literature method,³² was suspended in dry THF (150 mL) and stirred under reflux with potassium (1.56 g, 0.04 mol). The solution was stirred until the potassium was no longer visible, then iodoethane (7.70 g, 0.05 mol) was added. After 1 h, the solvent was concentrated and the resulting powder was washed with distilled water several times to remove soluble product (KI). The undissolved substances were recrystallised with methanol and the ligand was deposited as a pale-yellow powder; m.p. 170–173 °C; Anal. calcd for C₂₀H₂₂N₄S: C, 67.61; H, 6.19; N, 15.77; found: C, 67.78; H, 6.03; N, 15.63%; ¹H NMR (400 MHz, CDCl₃) δ : 7.31 (m, 4H, Ph), 4.12 (m, 2H, CH₂), 4.28 (s, 2H, SCH₂), 1.49 (m, 3H, CH₃); IR (ν_{max} /cm^{–1}) (KBr): 1274 $\nu_{\text{(C–N)}}$, 1458 $\nu_{\text{(C=N)}}$, 1616 $\nu_{\text{(C=C)}}$; UV-Vis (λ , nm, DMF): λ = 270, 280; Λ_{M} (DMF, 297 K): 2.9 S·cm²·mol^{–1}.

Complex [Ag(bebt)cinnamate] (2): To a stirred solution of ligand **1** (0.035 g, 0.10 mmol) in EtOH (5 mL) was added silver cinnamate (0.0255 g, 0.1 mmol) in EtOH (5 mL). The resultant solution was allowed to evaporate slowly at room temperature in the dark. Block



Scheme 1 The synthetic route to ligand bebt (1).

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crystals suitable for X-ray diffraction studies were obtained after two weeks; 0.048 g, 80% yield; Anal. calcd for $C_{29}H_{29}AgN_4O_2S$: C, 57.52; H, 4.79; N, 9.26; O, 5.29; S, 5.29; Ag, 17.85; found: C, 57.52; H, 4.83; N, 9.25; O, 5.28; S, 5.30; Ag, 17.81%; IR ($\nu_{\max}$ /cm⁻¹) (KBr): 1180 $\nu_{(C-N)}$, 1430 $\nu_{(C=N)}$, 1644 $\nu_{(C=C)}$; UV-Vis (λ , nm, DMF): $\lambda = 269, 282$; Λ_M (DMF, 297 K): 3.7 S·cm²·mol⁻¹.

X-Ray crystal structure determination

A suitable single crystal was mounted on a glass fibre and the intensity data were collected with a Bruker APEX II area detector with graphite-monochromated Mo-K α radiation ($\lambda = 0.71073$ Å) at 296 K. Cell refinement and data reduction were performed with the SAINT suite of programs.³³ The absorption corrections were made by empirical methods. The structure was solved by direct methods and refined by full-matrix least-squares against F^2 using SHELXTL software.³⁴ The non-H atoms in the structure were subjected to anisotropic refinement. Hydrogen atoms were located geometrically and treated with the riding model. The crystal data and experimental parameters relevant to the structure determination are listed in Table 1. Selected bond distances and angles are shown in Table 2.

DNA-binding studies

The application of electronic absorption spectroscopy is one of the most useful techniques in DNA-binding studies. A compound binding to DNA through intercalation is characterised by hypochromism in absorbance and red shift in wavelength, due to the intercalation mode involving a strong stacking interaction between the aromatic chromophore and the DNA base pairs. The magnitudes of the red shift

and the degree of hypochromism are furthermore commonly found to correlate with the strength of the intercalative interaction.^{35–38} In order to compare quantitatively the binding strength of the compounds, the intrinsic binding constants, K_b , of the compounds with DNA were obtained by monitoring the changes in absorbance at 270–280 nm for the compounds with increasing concentration of DNA. The absorption spectral titrations were performed by increasing an equal amount of DNA (5 μ L) to the compounds (25 μ L) and the reference solution in 2.5 mL Tris-HCl buffer to eliminate the absorption of DNA itself. Each sample solution was scanned in the range of 200–500 nm. The constant (K_b) was obtained by the following equation:³⁹

$$[DNA]/(\epsilon_a - \epsilon_f) = [DNA]/(\epsilon_b - \epsilon_f) + 1/K_b(\epsilon_b - \epsilon_f)$$

where [DNA] is the concentration of CT-DNA in the base pairs, ϵ_a corresponds to the observed extinction coefficient ($A_{\text{obsd}}/[M]$), ϵ_f corresponds to the extinction coefficient of the free compound, ϵ_b is the extinction coefficient of the compound when fully bound to CT-DNA and K_b is the intrinsic binding constant. The ratio of slope to the intercept in the plot of $[DNA]/(\epsilon_a - \epsilon_f)$ versus [DNA] gave the value of K_b .^{40,41}

No luminescence was observed for the two complexes at room temperature in any organic solvent or in the presence of CT-DNA. The fluorescence intensity of ethidium bromide (EB) is very weak in solvents, but it is greatly increased when EB is specifically intercalated into the base pairs of double stranded DNA and the enhanced fluorescence of EB can be quenched by the addition of a second molecule.⁴² To further clarify the interaction of the complex with DNA, a competitive binding experiment was carried out in a buffer by keeping $[DNA]/[EB] = 1$ and varying the concentrations of the complex. The fluorescence spectra of EB were measured using an excitation wavelength at 520 nm. The emission range was set between 550 and 750 nm. The spectra were analysed according to the classical Stern–Volmer equation:^{43,44}

$$I_0/I = 1 + K_{sv}$$

where I_0 and I are the fluorescence intensities in the absence and presence of the quencher, respectively, and $[Q]$ is the concentration of the complex.

Photophysical studies have proven quite useful in determining the binding modes of metal complexes to DNA, but do not give sufficient clues to support a binding mode.⁴⁵ Hydrodynamic measurements that are sensitive to the changes in DNA length are regarded as the least ambiguous and most critical tests of a binding model in solution in the absence of crystallographic data.⁴⁶ Viscosity experiments were conducted on an Ubbelohde viscometer, immersed in a water bath maintained at 25.0 ± 0.1 °C. Titrations were performed for the complex (3 μ M) and each compound was introduced into a CT-DNA solution (50 μ M) present in the viscometer. Data were analysed as $(\eta/\eta_0)^{1/3}$ versus the ratio of the concentration of the compound to CT-DNA, where η is the viscosity of CT-DNA in the presence of the compound and η_0 is the viscosity of CT-DNA alone. Viscosity values were calculated from the observed flow times of CT-DNA containing solutions corrected for the flow time of buffer alone (t_0), $\eta = (t - t_0)$.⁴⁷

Results and discussion

The complex **2** was prepared by reaction of silver cinnamate with bebt in ethanol. It is very soluble in DMF and DMSO, slightly soluble in water, ethanol, methanol and insoluble in organic solvents such as acetone, petroleum *etc.* The results of the elemental analyses show that the composition of the complex **2** is $[Ag(\text{bebt})\text{cinnamate}]$. The molar conductance values show that the complex is a non-electrolyte in DMF.⁴⁸

IR and UV-Vis spectra

The IR spectrum of the ligand **1** shows characteristic absorption bands of the benzimidazole group at 1274 and 1458 cm⁻¹, which

Table 1 Crystal data and structure refinement for complex **2**

Complex	2
Molecular formula	$C_{29}H_{29}AgN_4O_2S$
Molecular weight	605.49
Crystal size/mm ³	0.40 × 0.38 × 0.30
Crystal system	Triclinic
Space group	P-1
$a/\text{\AA}$	9.342(2)
$b/\text{\AA}$	12.847(3)
$c/\text{\AA}$	14.461(3)
$\alpha/^\circ$	90.853(3)
$\beta/^\circ$	108.176(3)
$\gamma/^\circ$	101.607(2)
$V/\text{\AA}^3$	1609.7(6)
Z	2
T/K	296(2)
$D_{\text{calcd}}/\text{g cm}^{-3}$	1.249
$F(000)/e$	620
θ range for data collection, deg	1.49–25.09
hkl range	–11,11/ 15,15/–13,17
Reflections collected	10773
Independent reflections / R_{int}	5597 / 0.0301
Data / restraints / parameters	5597 / 51 / 336
Final R_1 / wR_2 indices [$I > 2\sigma(I)$]	0.0593/0.1745
R_1 / wR_2 indices (all data)	0.0849/0.1900
Goodness-of-fit on F^2	1.021
Largest diff. peak / hole / e $\text{\AA}^{-3}$	1.656 / –1.115

Table 2 Selected bond lengths (Å) and bond angles (°) of complex **2**

Table 2. Selected bond lengths (Å) and bond angles (°) for complex 2			
Bond lengths		/Å	
Ag(1)–N(2)	2.227(5)	Ag(1)–O(1)	2.326(5)
Ag(1)–N(3)	2.283(4)		
Bond angles		/°	
N(2)–Ag(1)–N(3)	107.81(16)	N(3)–Ag(1)–O(1)	112.34(17)
N(2)–Ag(1)–O(1)	139.00(18)		

can be assigned to $\nu_{(\text{C}=\text{N})}$ and $\nu_{(\text{C}=\text{N})}$, respectively.^{49,50} For the complex **2**, these stretching frequencies are red shifted to 1180 cm^{-1} and 1430 cm^{-1} , respectively, which implies direct coordination of nitrogen atoms to metal ions.

The electronic spectra of the ligand **1** and the complex **2** were recorded in DMF solution at room temperature. The UV-Vis bands of ligand **1** (270, 280), which are marginally shifted in complex **2** (269, 282), are assigned to $\pi \rightarrow \pi^*$ (benzene) and $\pi \rightarrow \pi^*$ (imidazole) transitions, which show clear evidence of (C=N–C=C) coordination to the metal ion centre.⁵¹

X-Ray crystallography

The complex **2** crystallises in the triclinic space group P-1 and its structure, along with the atomic numbering scheme, is shown in Fig. 1. The complex **2** has a mononuclear three-coordinate configuration, consisting of a central metal Ag(I) atom, plus bebt and cinnamate ligands. The Ag(I) atom is coordinated by one oxygen atom from a cinnamate anion and two nitrogen atoms from a bebt ligand. The coordination geometry around Ag(I) is best described as trigonal planar. In addition, the

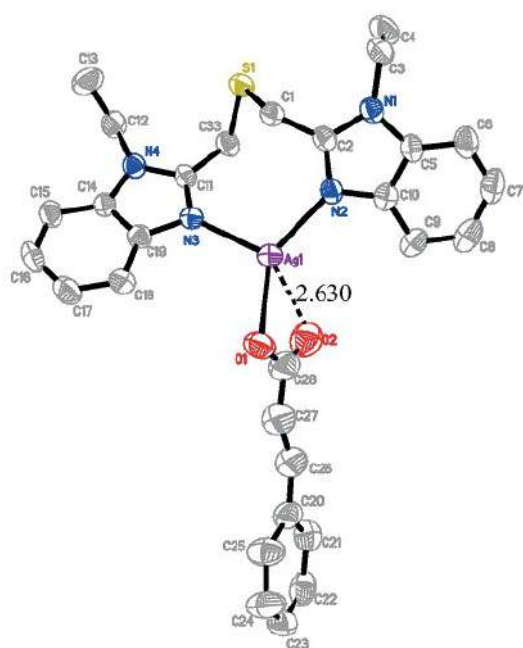


Fig. 1 Molecular structure of complex **2** with displacement ellipsoids drawn at the 30% probability level; H atoms are omitted for clarity.

cinnamate group shows weak coordination to Ag(I) in the chelated mode with an Ag–O(2) distance of 2.630(5) Å as shown in Fig. 1.

As one of the important types of supramolecular forces, $\pi - \pi$ stacking shows a specific structural requirement for substrate recognition or the arrangement of complicated architectures. The crystal packing of Ag(I) complex is shown in Fig. 2. Four adjoining complexes are stabilised by weak $\pi - \pi$ stacking of the benzimidazole rings with the vertical distances 3.632 Å and 3.784 Å.

CT–DNA binding studies

Electronic absorption titration

The electronic absorption spectra of free ligand **1** and complex **2** in the absence and presence of CT–DNA are given in Fig. 3. As Fig. 3 shows, there is a band at 276 nm for the free ligand **1**, while there is a band at 270 nm for complex **2**. Upon addition of increasing CT–DNA concentrations, these bands exhibited hypochromism of 10.7 and 29.84%, respectively. From the electronic absorption spectroscopy experiments, K_b values of **1** and complex **2** were obtained as $1.22 \times 10^4 \text{ M}^{-1}$ ($R^2 = 0.8691$ for 9 points) and $3.02 \times 10^4 \text{ M}^{-1}$ ($R^2 = 0.9847$ for 10 points) respectively. Compared with those of so-called DNA-intercalative complexes ($K_b = 2.9 \times 10^4 - 1.2 \times 10^5 \text{ M}^{-1}$),^{41,52} suggests that both the ligand and complex can bind to DNA *via* an intercalation mode. Hence, the order of the binding affinity is **2** > **1**.

In general, this phenomenon may be attributed to variation of electron density. The charge transfer of coordinated ligand **1** resulting from coordination to the Ag(I) atom should reduce the charge density of the benzimidazole, which is conducive to intercalation. Thus the affinity for DNA is stronger in the case of complex **2** when compared with the ligand **1**.

Competitive binding with EB

Competitive EB binding studies were undertaken in order to examine the binding of each complex with DNA. EB (ethidium bromide) is a conjugate planar molecule. If a complex can replace EB from DNA-bound EB, the fluorescence of the solution will be quenched due to the fact that free EB molecules are readily quenched by surrounding water molecules.⁵³ The Stern–Volmer plots DNA–EB (Fig. 4) show that the quenching of EB bound to DNA by the two compounds follows a linear relationship, consistent with intercalation of the test compounds. The K_{sv} values for free bebt and complex were $2.49 \times 10^3 \text{ M}^{-1}$

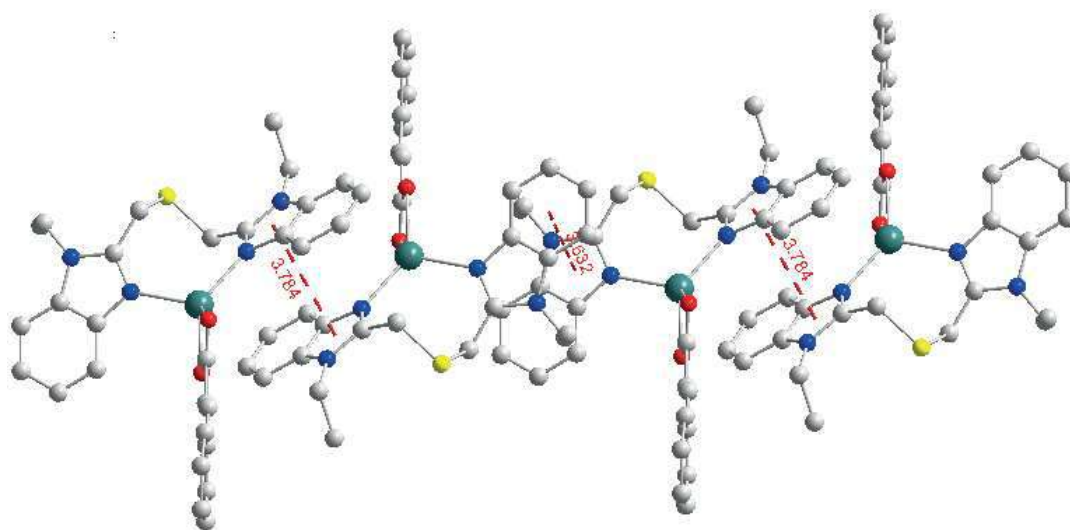


Fig. 2 The structure of the complex linked *via* $\pi - \pi$ stacking interaction.

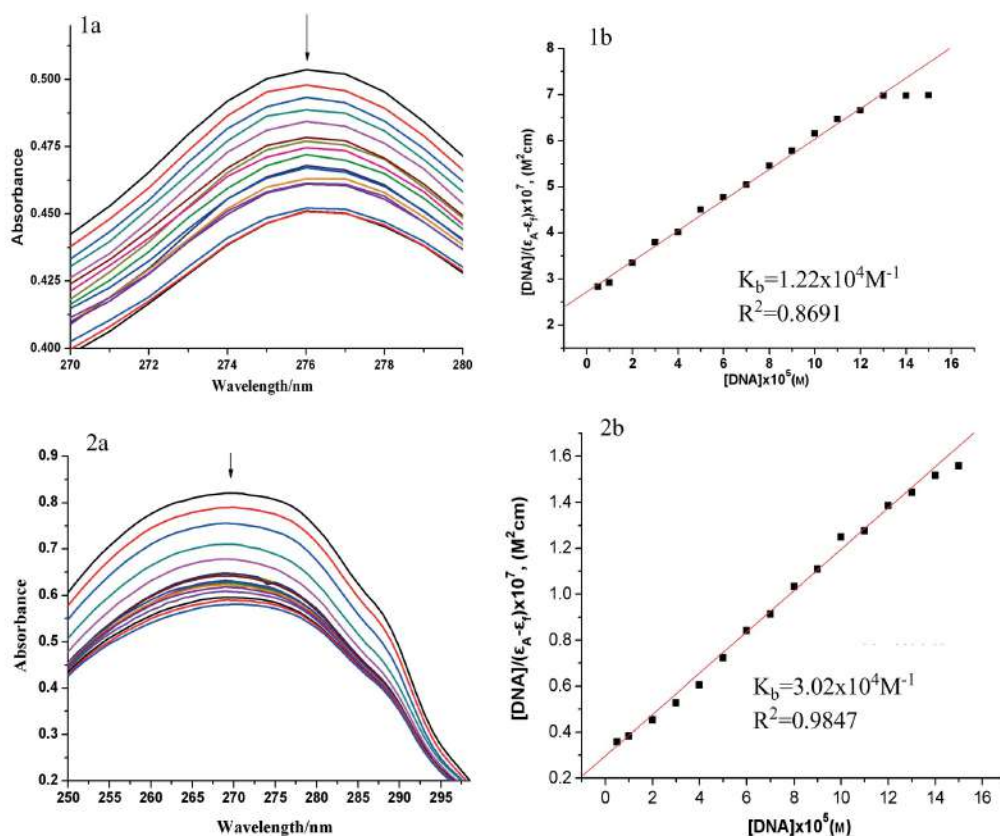


Fig. 3 Electronic spectra of free ligand **1** (1a), complex **2** (2a) in Tris-HCl buffer upon addition of CT-DNA. The arrow shows the emission intensity changes upon increasing DNA concentrations. $[DNA] / (\epsilon_A - \epsilon_f)$ versus $[DNA]$ for the titration of the ligand **1** (1b), the complex **2** (2b) with CT-DNA.

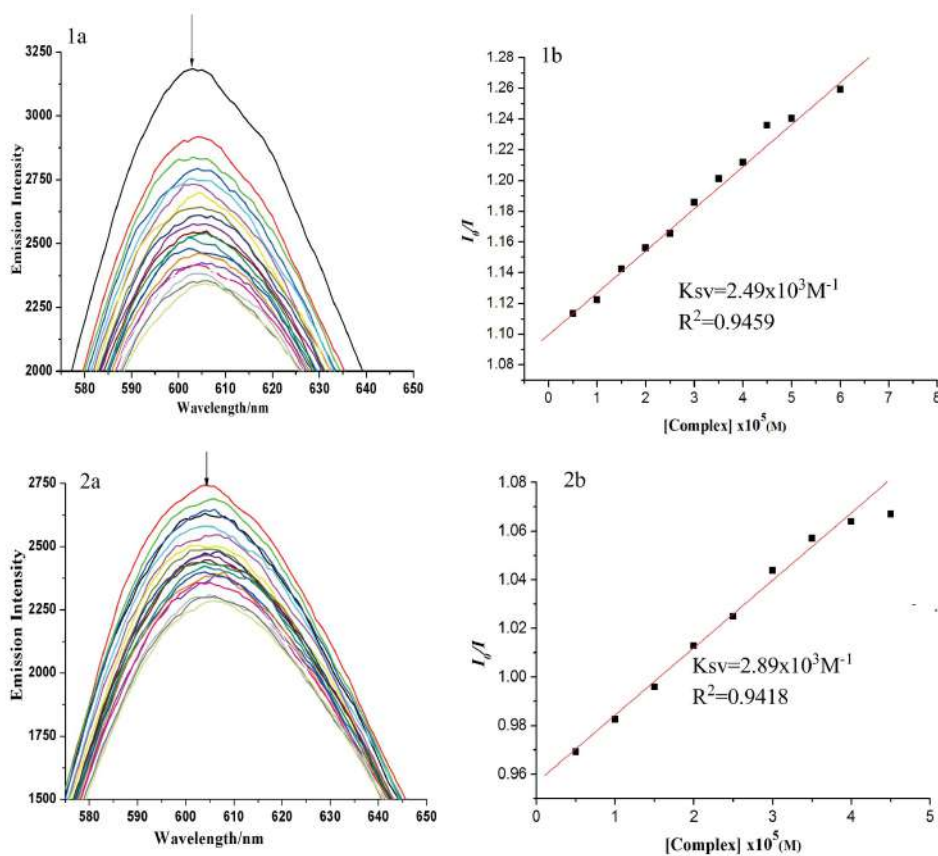


Fig. 4 Emission spectra of EB bound to CT-DNA in the presence of ligand **1** (1a) and complex **2** (2a), $\lambda_{exc} = 520 \text{ nm}$. The arrows show the intensity changes upon increasing concentrations of the complexes. Fluorescence quenching curves of EB bound to CT-DNA by the free ligand **1** (1b) and the complex **2** (2b). (Plots of I_0/I versus $[Complex]$).

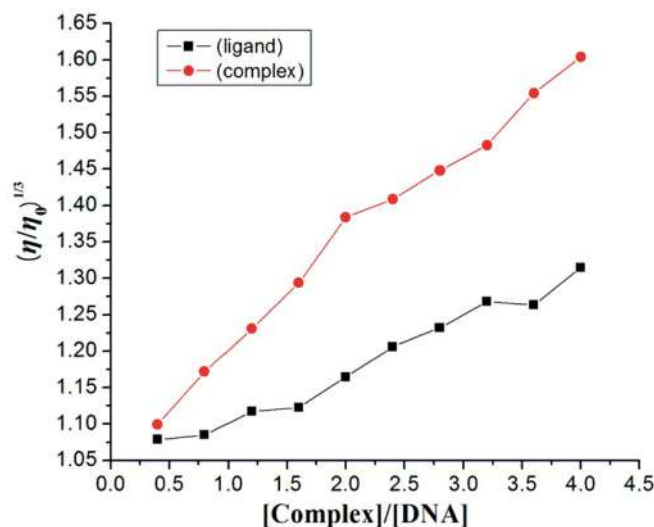


Fig. 5 Effect of increasing amounts of the compounds on the relative viscosity of DNA at 25.0 ± 0.1 °C.

($R^2 = 0.9459$) and $2.89 \times 10^3 \text{ M}^{-1}$ ($R^2 = 0.9418$), respectively. The phenomena suggest that the complex can compete for DNA-binding sites with EB and displace EB from the EB–DNA system,⁵⁴ which is usually characteristic of the intercalative interaction of compounds with DNA.⁵⁵

Viscosity studies

Viscosity measurements were carried out on CT–DNA by varying the concentration of the complex. Figure 5 shows that ligand **1** and complex **2** cause increase in the relative viscosity of DNA. This may be explained by the fact that the compounds can intercalate the adjacent DNA base pairs, leading to an increase in the separation of base pairs at intercalation sites, thus, an increase in overall DNA contour length.⁵⁶ The results above demonstrate that the ligand and its metal complex could bind to DNA by an intercalation mode.

Conclusions

1,3-Bis(1-ethylbenzimidazol-2-yl)-2-thiopropane **1** and its Ag(I) complex **2** have been synthesised and characterised. The interaction of compounds **1** and **2** with CT–DNA has been explored by using absorption and fluorescence spectral techniques and viscosity measurements. The data suggest that both compounds bind to CT–DNA in an intercalation mode and complex **2** can bind to DNA more strongly than ligand **1**. These results should help to understand the mechanism of interactions of small molecules with DNA, so aiding the development of their biological, pharmaceutical and physiological applications.

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Electronic Supplementary Information

Crystallographic data for the Ag(I) complex has been deposited with the Cambridge Crystallographic Data Centre as supplementary publication No. CCDC 1402474. Copies of the data can be obtained free of charge on application to The Director, CCDC, 12 Union Road, Cambridge CB2 1EZ, UK (fax: +44 1223 336 033; E-mail: deposit@ccdc.cam.ac.uk).

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